

Comparison of abmneo and ABM - ABM part

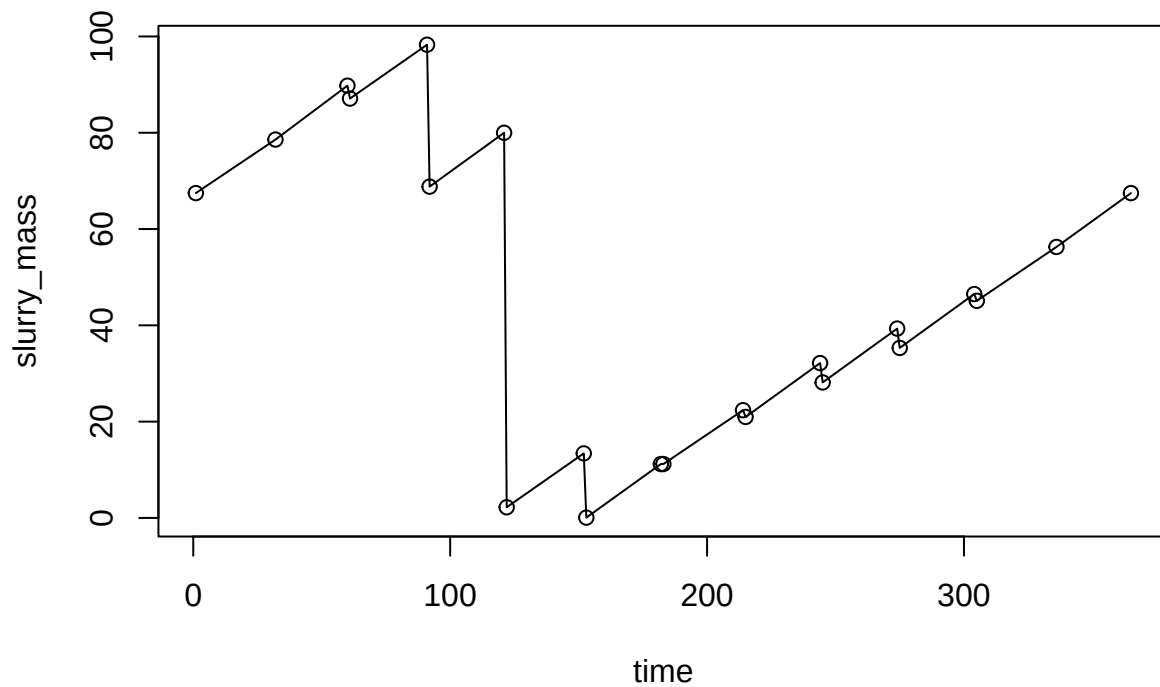
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08 May, 2026 10:09

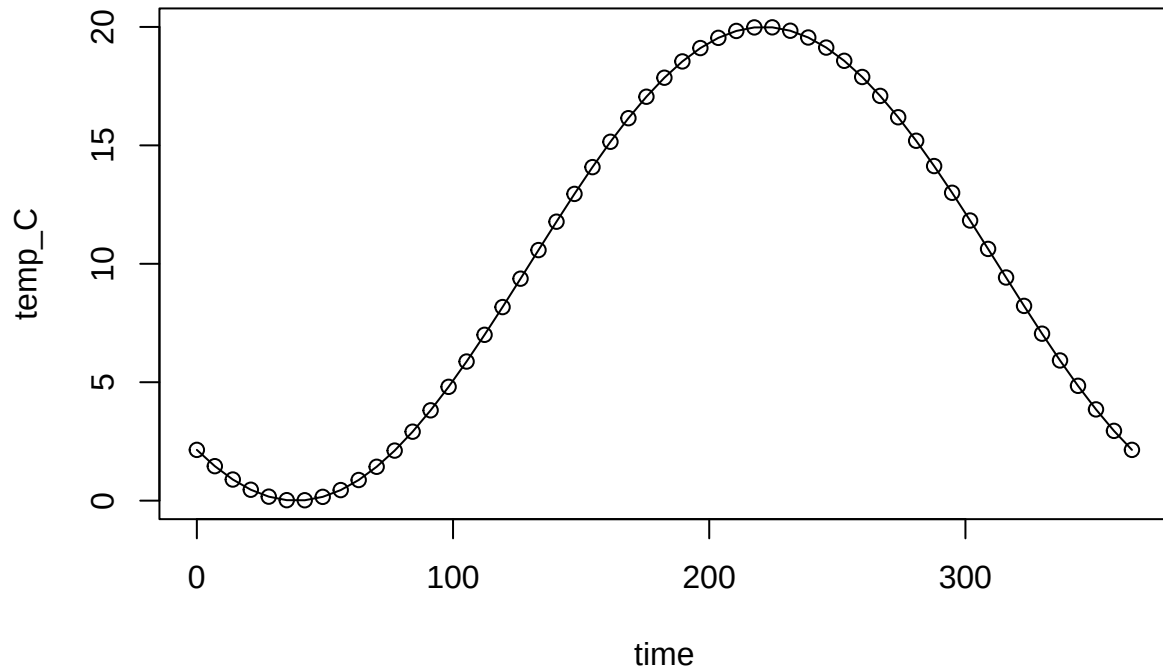
```
devtools::load_all('../..//ABM')
```

```
## i Loading ABM
```

```
mass_series <- read.csv('level_pig_od_ref.csv')  
plot(slurry_mass ~ time, data = mass_series, type = 'o')
```



```
temp_dat <- data.frame(time = seq(0, 365, length.out = 53))  
temp_dat$temp_C <- 10 + 10 * sin((temp_dat$time - 130) / 365 * 2 * pi)  
plot(temp_C ~ time, data = temp_dat, type = 'o')
```



```

arrh_pars <- list(lnA = 29.96298,
  E_CH4 = 81000,
  A = c(xa_dead= 3.61383 * 10^12, starch = 5 * 3.61383 * 10^12, Cfat = 0.1 * 3.61383 * 10^12),
  E = c(xa_dead= 81557, starch = 81557, Cfat = 81557, CPs = 81557, CPf = 81557, RFd = 81557),
  R = 8.314,
  VS_CH4 = 6.67,
  scale_alpha_opt = list(VSd = 0.46056, notVSd = 0.556886, CP = 2.065),
  kl = c(NH3 = 336, NH3_floor = 0.36*337, H2S = 0.02),
  scale_EF_NH3 = 0.0000048)

grp_pars <- list(grps = c('m0', 'm1', 'm2','m3','m4','m5','sr1'),
  yield = c(default = 0.05, sr1 = 0.065),
  xa_fresh = c(m0 = 0.06, m1 = 0.06, m2 = 0.06, m3 = 0.06, m4 = 0.06, m5 = 0.06, sr1 = 0),
  xa_init = c(all = 0.1),
  decay_rate = c(all = 0.02),
  ks_coefficient = c(default = 1.153337, sr1 = 0.461335),
  qhat_opt = c(m0 = 0.46289, m1 = 0.74568, m2 = 0.9006605, m3 = 2.79672, m4 = 4.66012, m5 = 55, sr1 = 43.75),
  T_opt = c(m0 = 18, m1 = 18, m2 = 28, m3 = 36, m4 = 43.75, m5 = 55, sr1 = 43.75),
  T_min = c(m0 = 0, m1 = 9.67, m2 = 9.67, m3 = 15, m4 = 26.25, m5 = 30, sr1 = 0),
  T_max = c(m0 = 25, m1 = 25, m2 = 38, m3 = 45, m4 = 51.25, m5 = 60, sr1 = 51.25),
  ki_NH3_min = c(all = 0.015),
  ki_NH3_max = c(all = 5.37),
  ki_NH4_min = c(all = 2.7),
  ki_NH4_max = c(all = 6.8492),
  ki_H2S_slope = c(default = -0.10623, sr1 = -0.1495),
  ki_H2S_int = c(default = 1.02, sr1 = 1.42),

```

```

ki_H2S_min = c(default = 0.08),
IC50_low = c(default = 0.20854, sr1 = 0.2772),
ki_HAC = c(default = 0.31),
pH_UL = c(default = 100),
pH_LL = c(default = -100))

```

Write mng_pars in order to insert slurry_mass data

```

mng_pars <- list(slurry_prod_rate = 5700,
  slurry_mass = mass_series,
  storage_depth = 2,
  resid_depth = 0,
  floor_area = 0,
  area = 0.03,
  empty_int = 0,
  temp_C = temp_dat,
  wash_water = 0,
  wash_int = NA,
  rest_d = 0,
  cover = 'none',
  resid_enrich = 0.9,
  graze = c(start = 'May', duration = 0, hours_day = 0),
  scale = c(ks_coefficient = 1, qhat_opt = 1, xa_fresh = 1, yield = 1, alpha_opt = 1)
)

```

Take man_pars from man_pars_pig2.0

```

man_pars_pig2.0 <- list(conc_fresh = list(sulfide = 0.01, urea = 3.17, sulfate = 0.01, TAN = 0.0, starch = 0.0,
  VFA = 1.7, xa_dead = 0, Cfat = 27.6, CPs = 21.1 * 0.55, CPf = 21.1,
  ash = 15), pH = 7, dens = 1000)

```

```

devtools::load_all('../..//ABM')

```

i Loading ABM

```

system.time(
  abm_out <- abm(
    days = 365,
    man_pars = man_pars_pig2.0,
    mng_pars = mng_pars,
    arrh_pars = arrh_pars,
    grp_pars = grp_pars,
    add_pars = list(evap = 0, rain = 0, COD_conv.CH4 = 5.32),
    resp = FALSE,
    pH_inhib_ouerrule = TRUE,
    startup = 2
  )
)

```

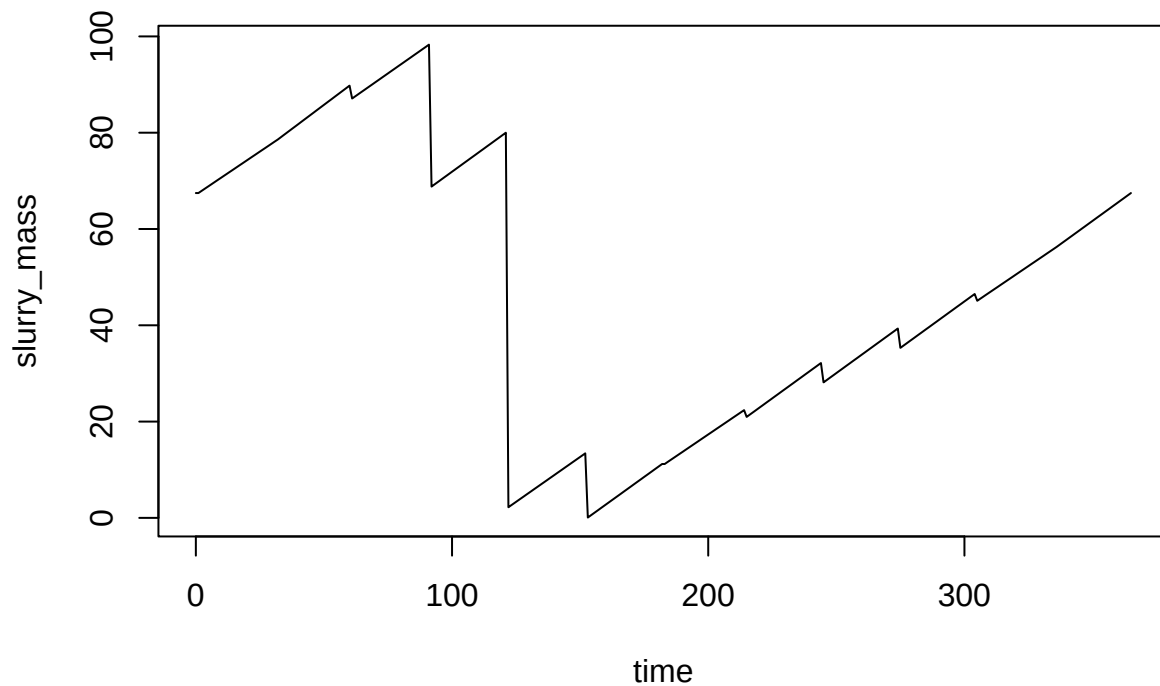
##

Startup run 1x ->

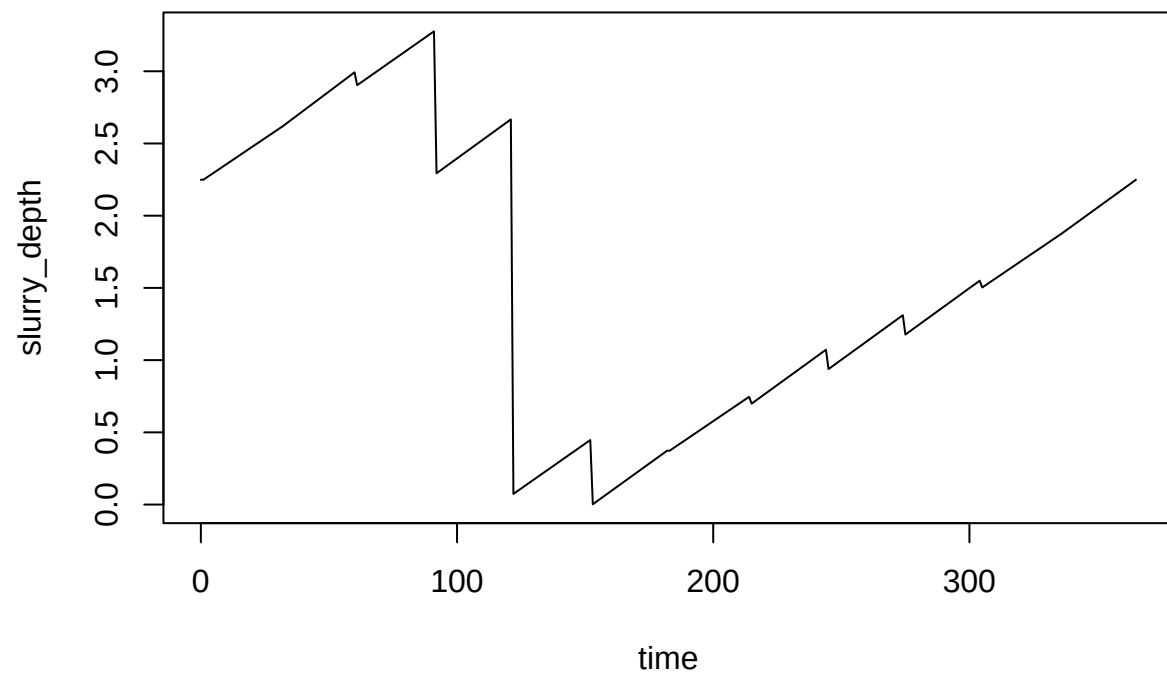
Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry mass reached
Check output.

arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
rain = 0 kg/m2/day

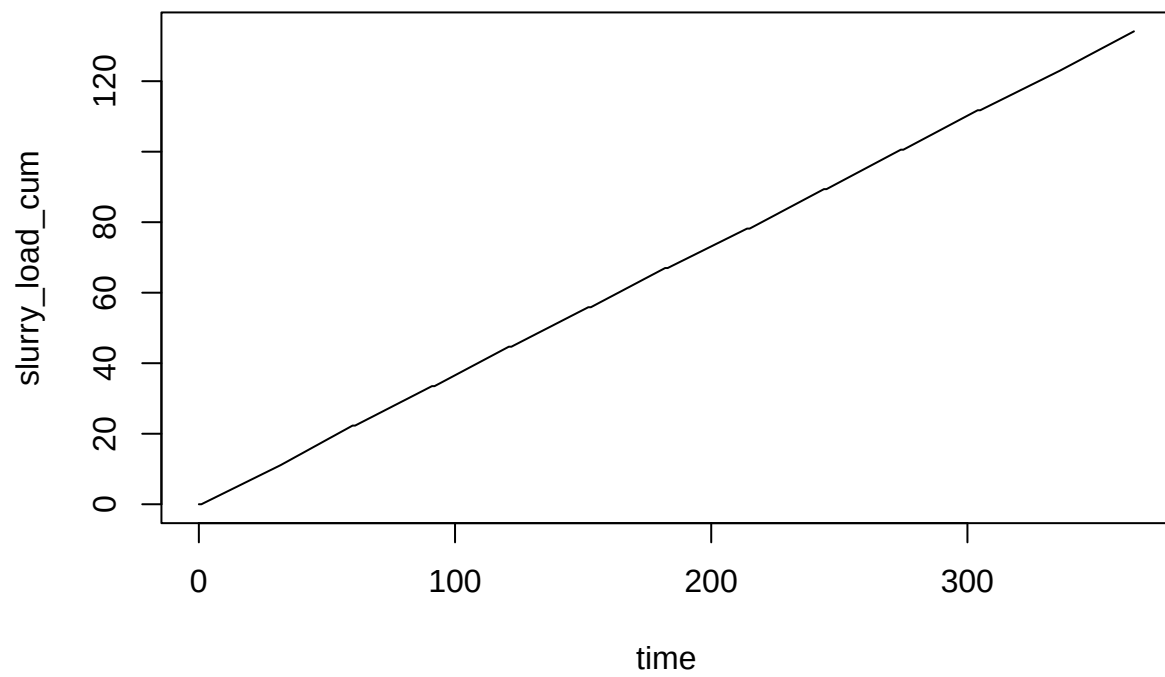
```
## evaporation = 0 kg/m2/day
## 2x ->
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
## Check output.
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day
## and final run
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
## Check output.
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
## rain = 0 kg/m2/day
## evaporation = 0 kg/m2/day
##      user  system elapsed
##    2.418   0.000   2.419
plot(slurry_mass ~ time, data = abm_out, type = 'l')
```



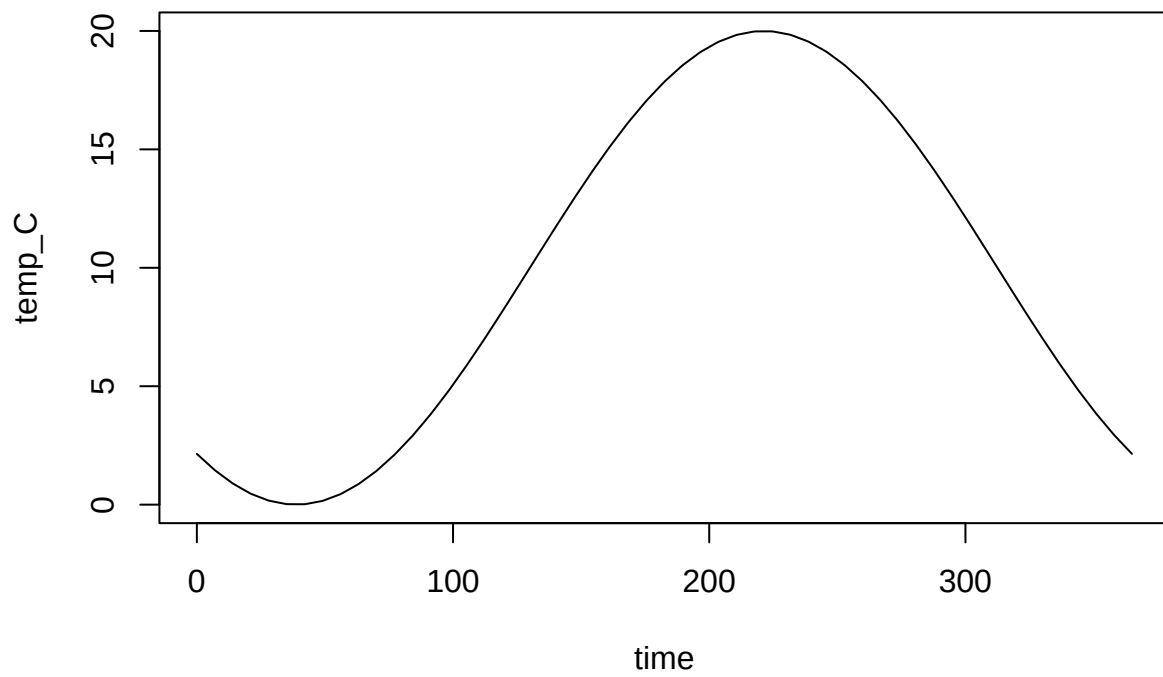
```
plot(slurry_depth ~ time, data = abm_out, type = 'l')
```



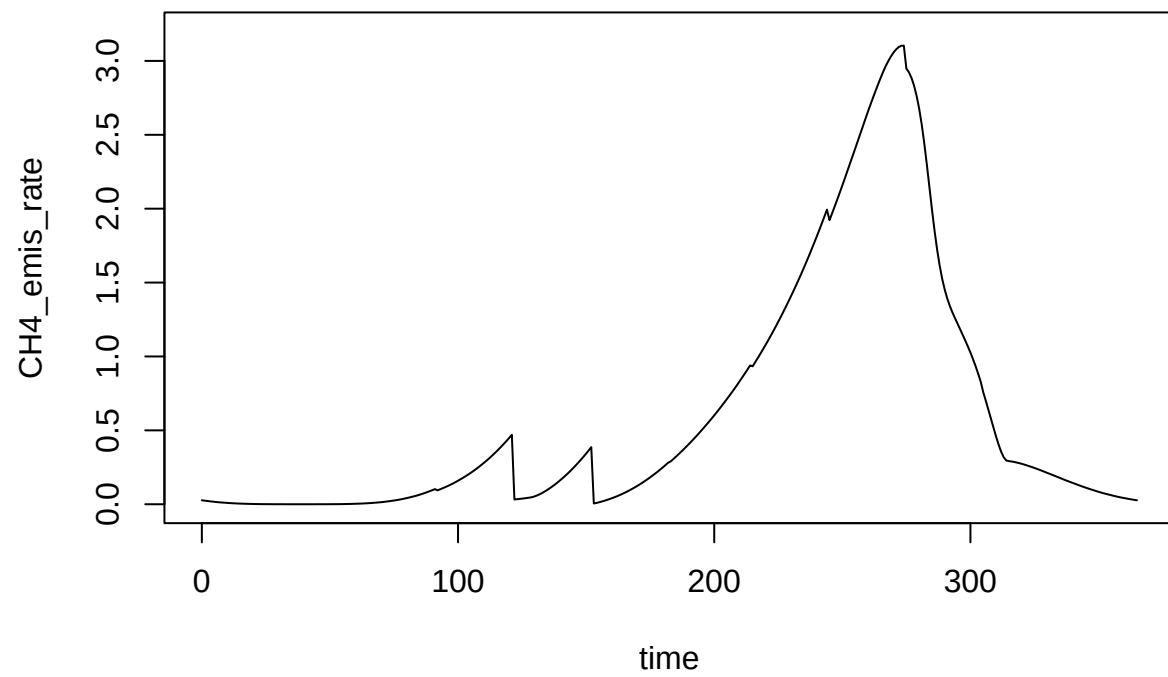
```
plot(slurry_load_cum ~ time, data = abm_out, type = 'l')
```



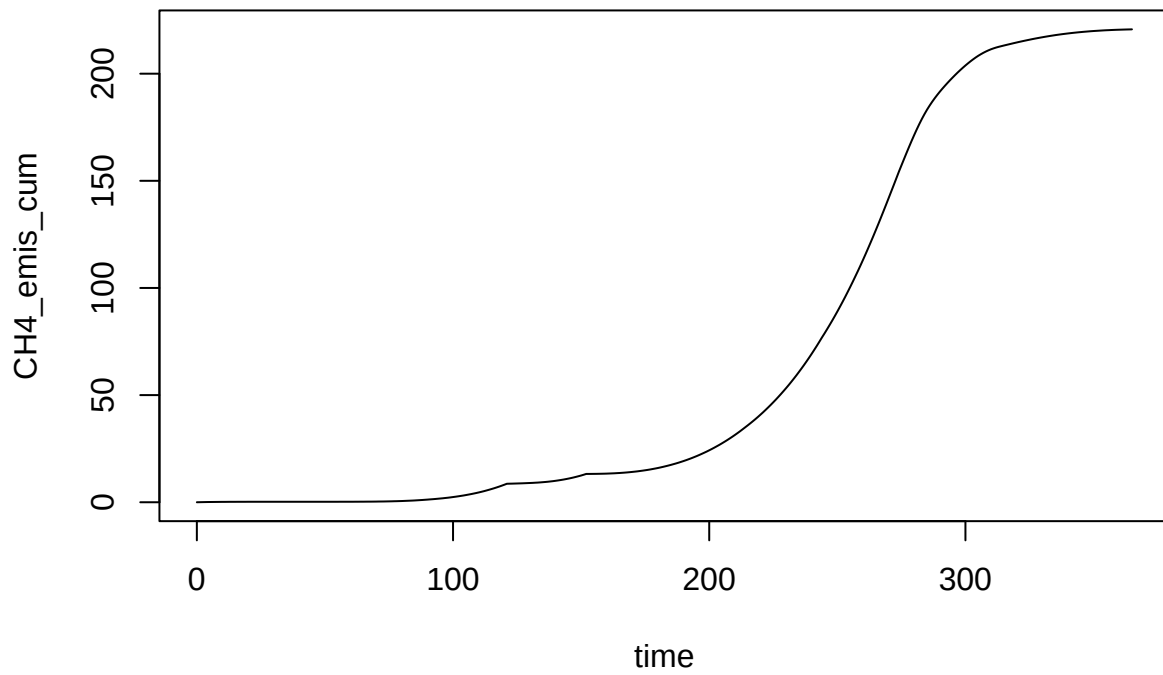
```
plot(temp_C ~ time, data = abm_out, type = 'l')
```



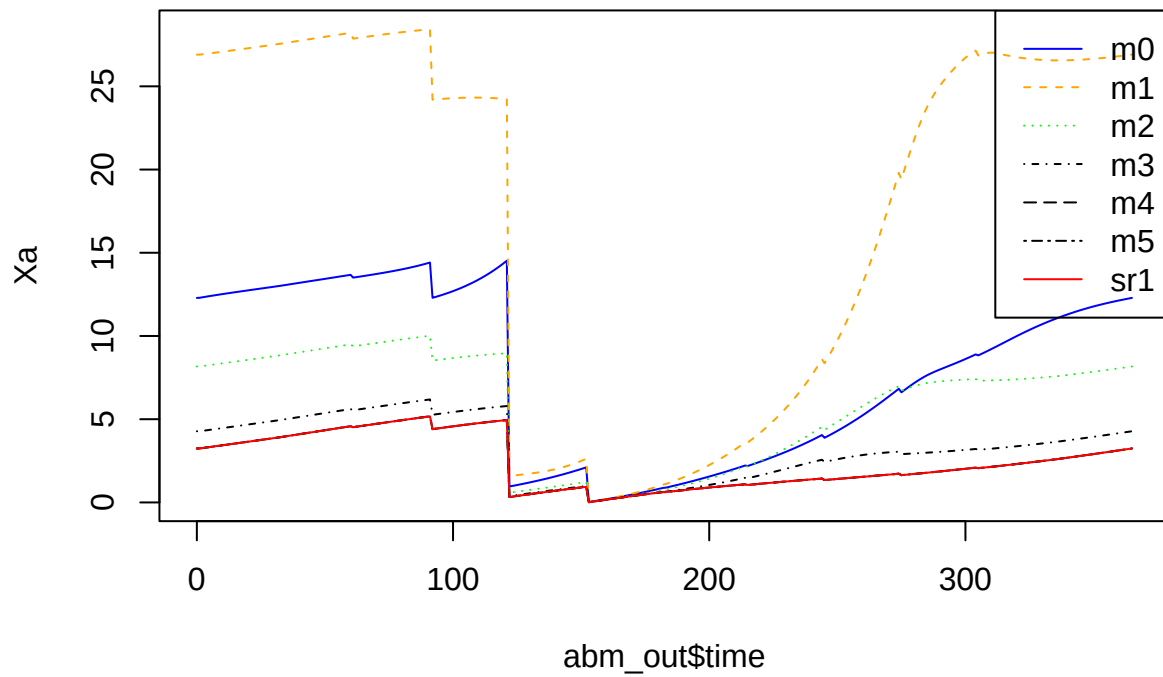
```
plot(CH4_emis_rate ~ time, data = abm_out, type = 'l', ylim = c(0, 3.2))
```



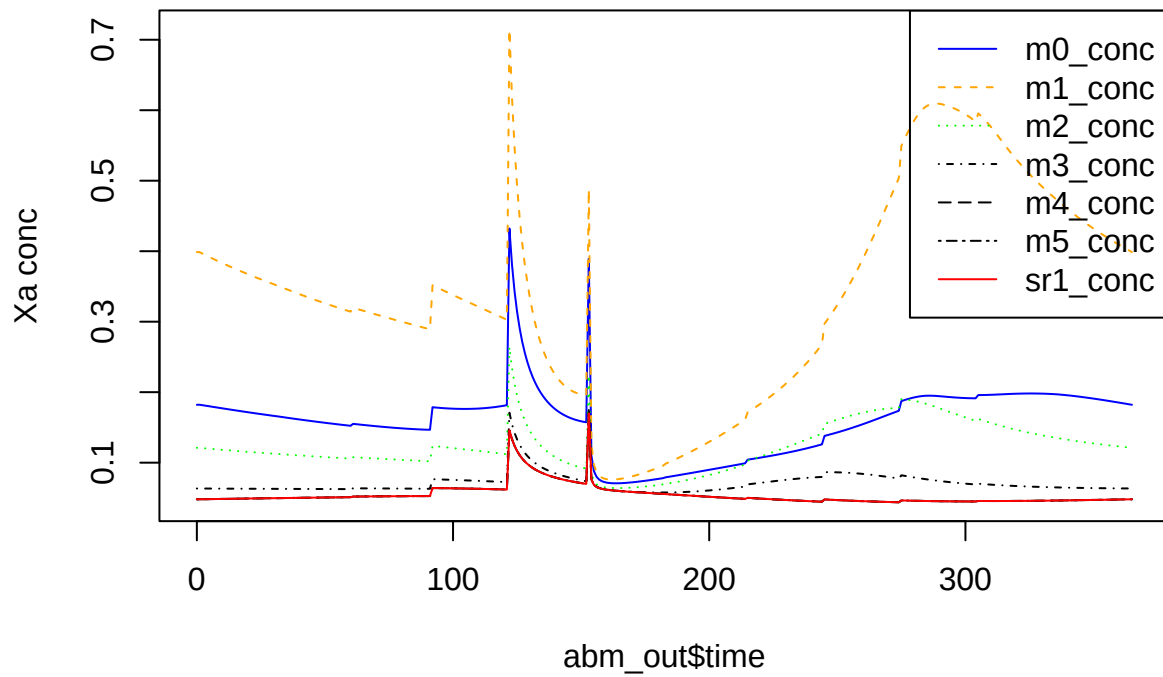
```
plot(CH4_emis_cum ~ time, data = abm_out, type = 'l')
```



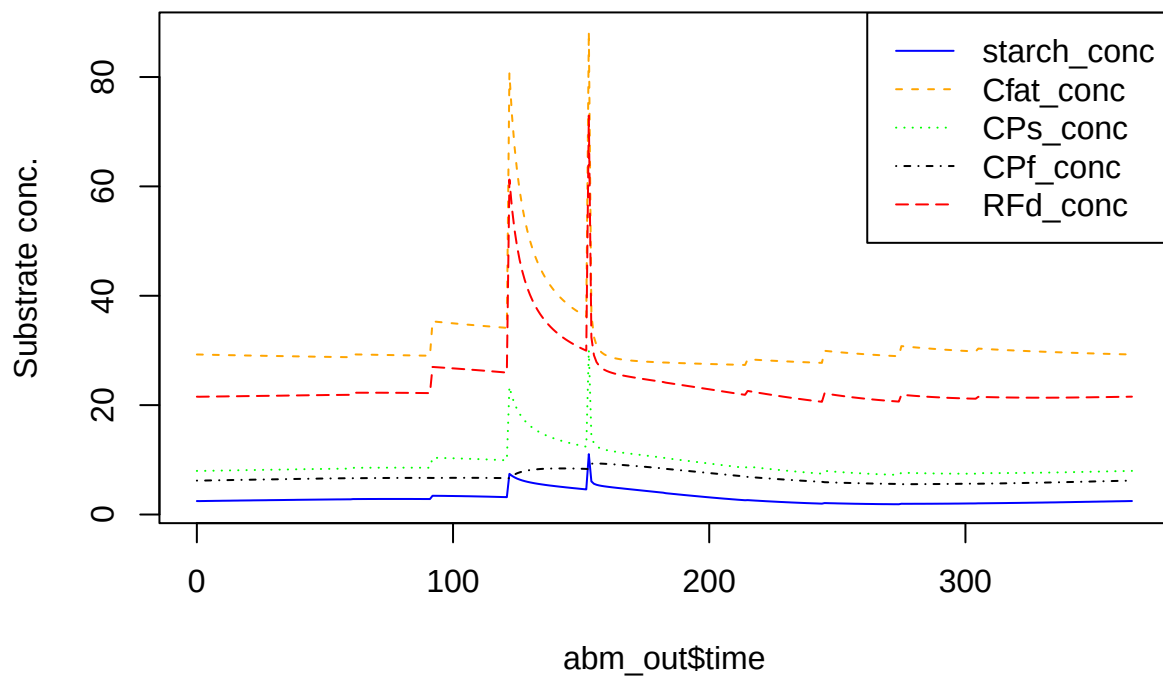
```
matplot(
  abm_out$time,
  abm_out[, nn <- grp_pars$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



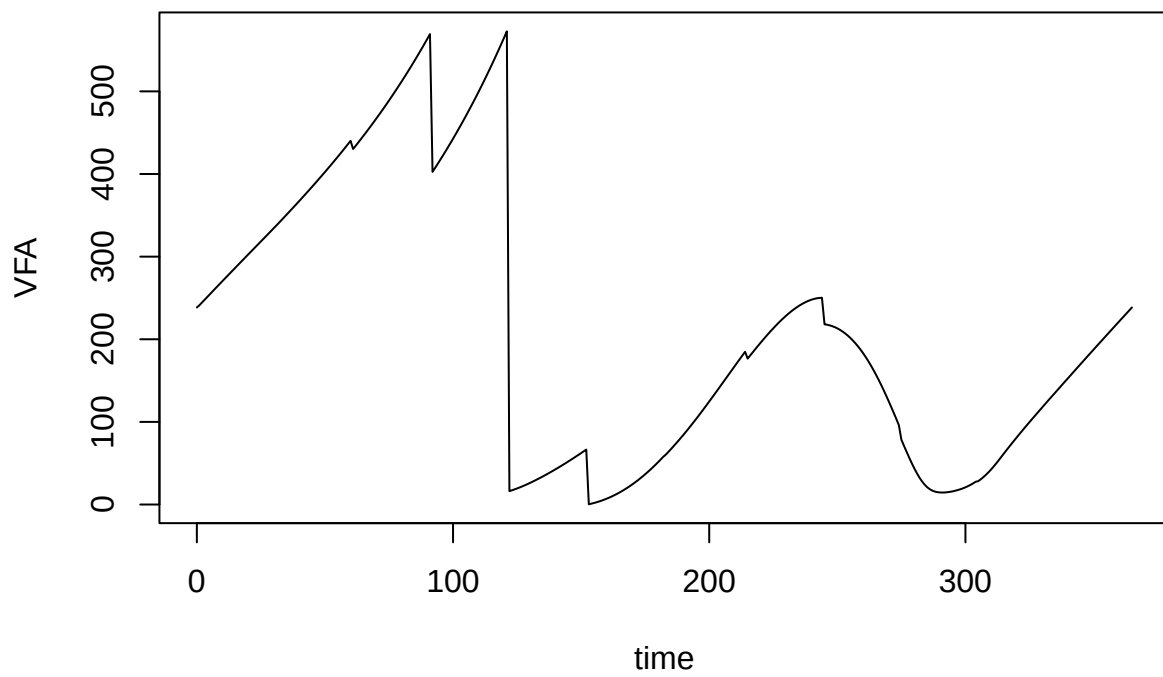
```
matplot(
  abm_out$time,
  abm_out[, nn <- paste0(grp_pars$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



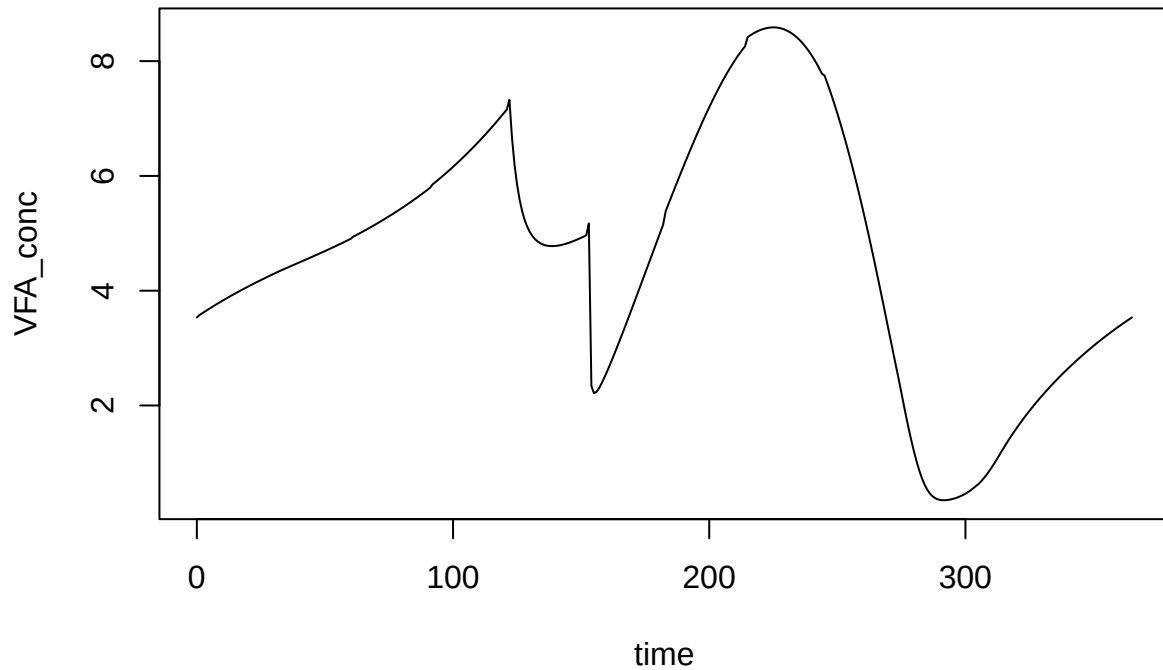
```
matplot(
  abm_out$time,
  abm_out[, nn <- paste0(c('starch', 'Cfat', 'CPs', 'CPf', 'RFd'), '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



```
plot(VFA ~ time, data = abm_out, type = 'l')
```



```
plot(VFA_conc ~ time, data = abm_out, type = 'l')
```



No methanogens, for comparison

```
grp_pars0 <- list(grps = c('m0', 'm1', 'm2', 'm3', 'm4', 'm5', 'sr1'),
  yield = c(default = 0.05, sr1 = 0.065),
  xa_fresh = c(all = 0),
  xa_init = c(all = 0),
  decay_rate = c(all = 0.02),
  ks_coefficient = c(default = 1.153337, sr1 = 0.461335),
  qhat_opt = c(m0 = 0.46289, m1 = 0.74568, m2 = 0.9006605, m3 = 2.79672, m4 = 4.66012, m5 = 5.5, sr1 = 43.75),
  T_opt = c(m0 = 18, m1 = 18, m2 = 28, m3 = 36, m4 = 43.75, m5 = 55, sr1 = 43.75),
  T_min = c(m0 = 0, m1 = 9.67, m2 = 9.67, m3 = 15, m4 = 26.25, m5 = 30, sr1 = 0),
  T_max = c(m0 = 25, m1 = 25, m2 = 38, m3 = 45, m4 = 51.25, m5 = 60, sr1 = 51.25),
  ki_NH3_min = c(all = 0.015),
  ki_NH3_max = c(all = 5.37),
  ki_NH4_min = c(all = 2.7),
  ki_NH4_max = c(all = 6.8492),
  ki_H2S_slope = c(default = -0.10623, sr1 = -0.1495),
  ki_H2S_int = c(default = 1.02, sr1 = 1.42),
  ki_H2S_min = c(default = 0.08),
  IC50_low = c(default = 0.20854, sr1 = 0.2772),
  ki_HAC = c(default = 0.31),
  pH_UL = c(default = 100),
  pH_LL = c(default = -100))
```

```
devtools::load_all(' ../../ABM')
```

```
## i Loading ABM
```

```
system.time(  
  abm_out0 <- abm(  
    days = 365,  
    man_pars = man_pars_pig2.0,  
    mng_pars = mng_pars,  
    arrh_pars = arrh_pars,  
    grp_pars = grp_pars0,  
    add_pars = list(evap = 0, rain = 0, COD_conv.CH4 = 5.32),  
    resp = FALSE,  
    pH_inhib_ouerrule = TRUE,  
    startup = 2  
  )  
)
```

```
##
```

```
## Startup run 1x ->
```

```
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
```

```
## Check output.
```

```
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
```

```
## rain = 0 kg/m2/day
```

```
## evaporation = 0 kg/m2/day
```

```
## 2x ->
```

```
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
```

```
## Check output.
```

```
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
```

```
## rain = 0 kg/m2/day
```

```
## evaporation = 0 kg/m2/day
```

```
## and final run
```

```
## Warning in abm(days = days, delta_t = delta_t, times = times, wthr_pars = wthr_pars, : Maximum slurry
```

```
## Check output.
```

```
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
```

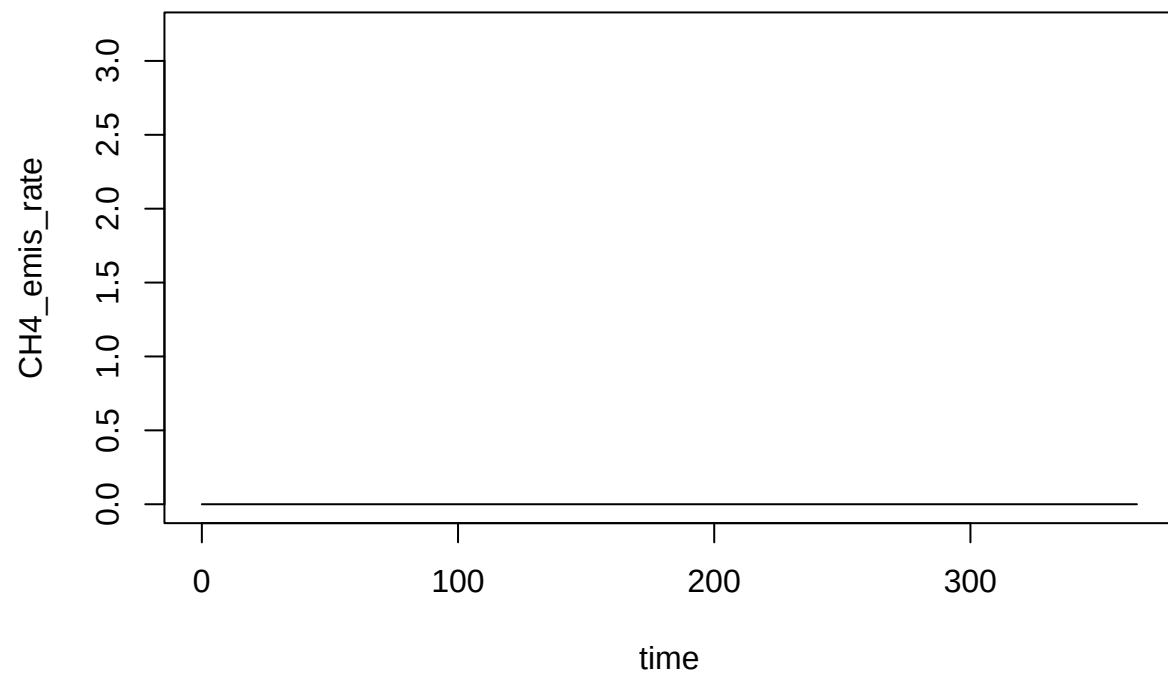
```
## rain = 0 kg/m2/day
```

```
## evaporation = 0 kg/m2/day
```

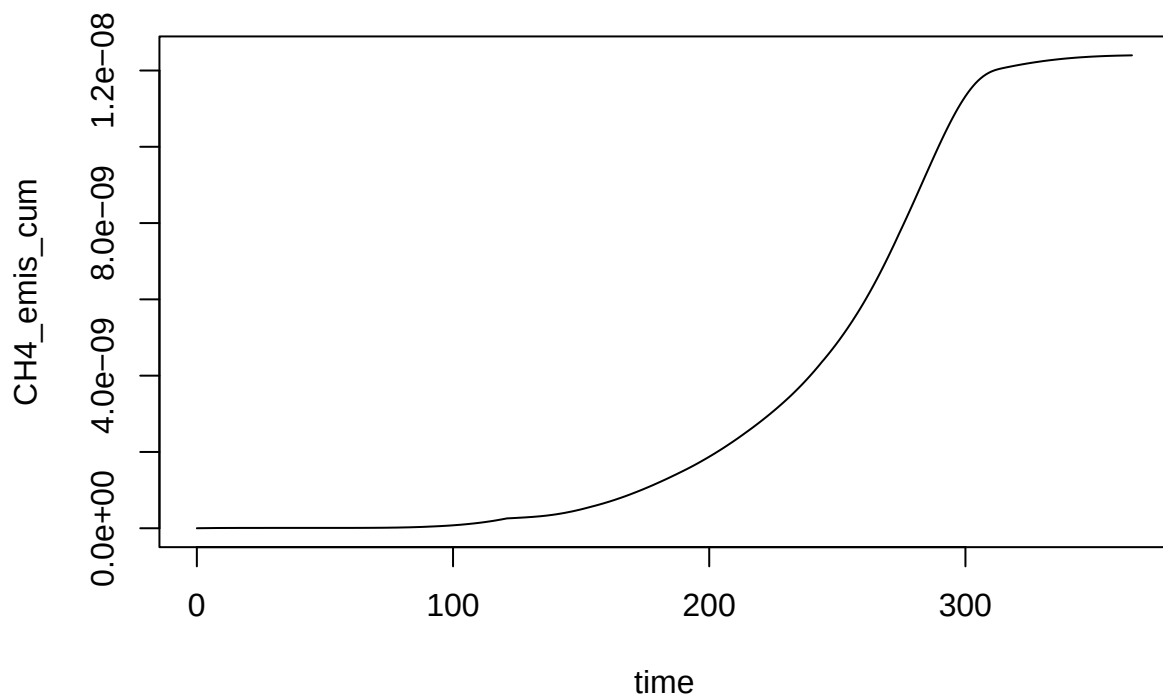
```
## user system elapsed
```

```
## 2.757 0.001 2.762
```

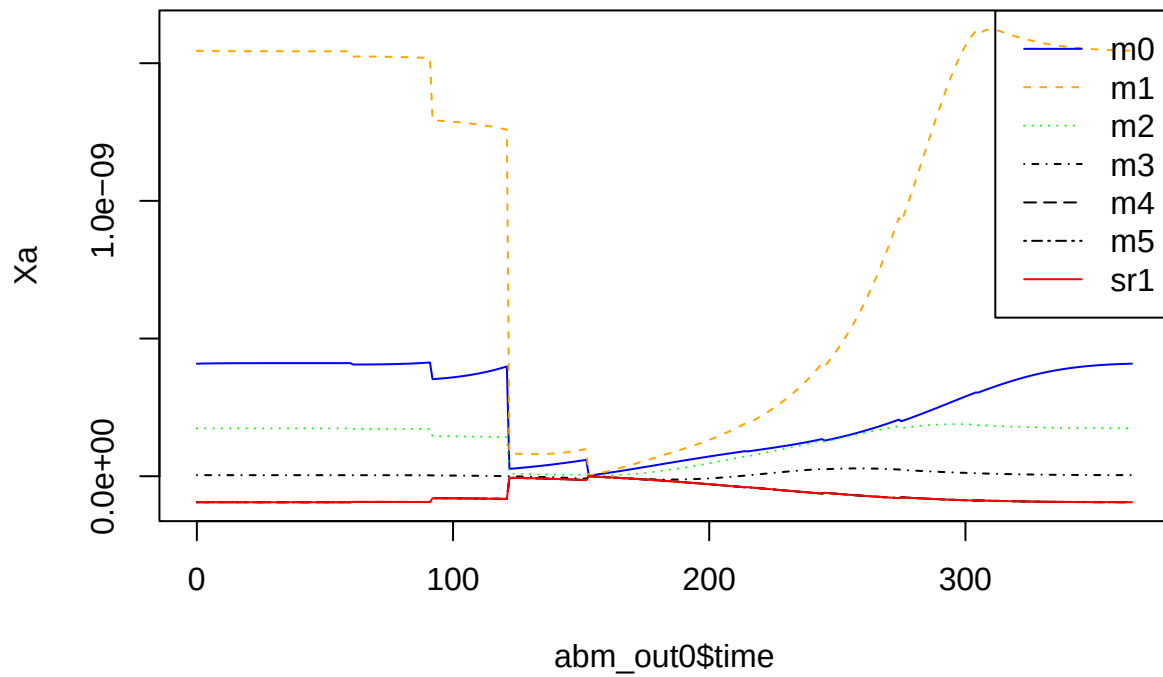
```
plot(CH4_emis_rate ~ time, data = abm_out0, type = 'l', ylim = c(0, 3.2))
```



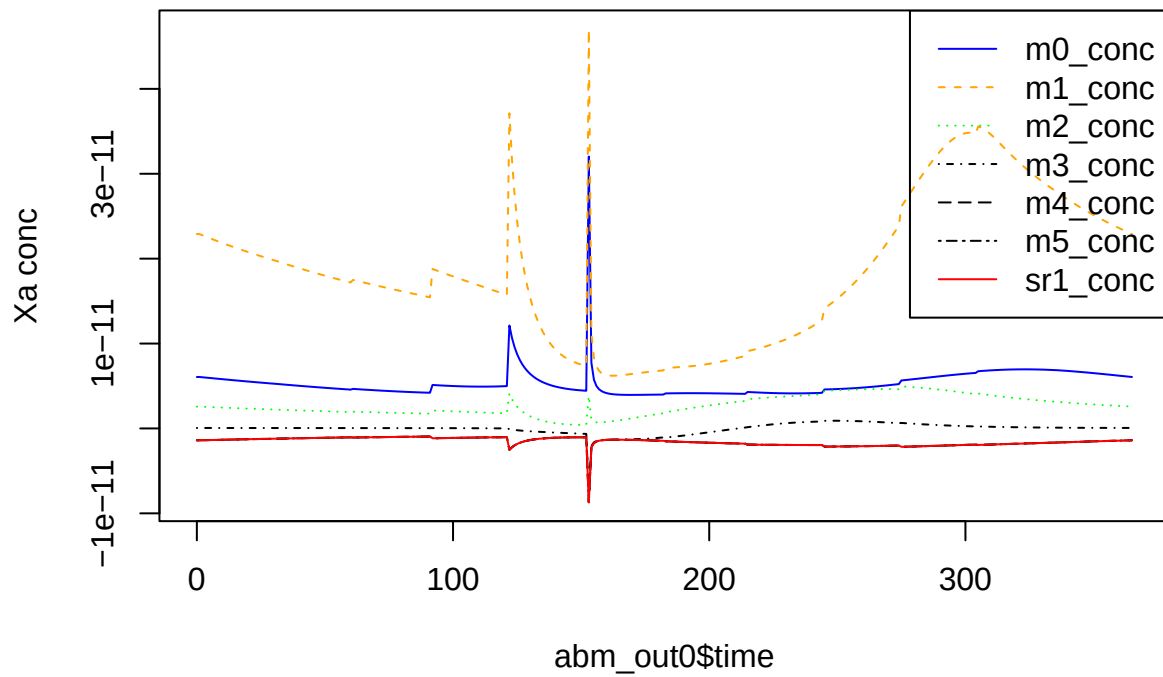
```
plot(CH4_emis_cum ~ time, data = abm_out0, type = 'l')
```



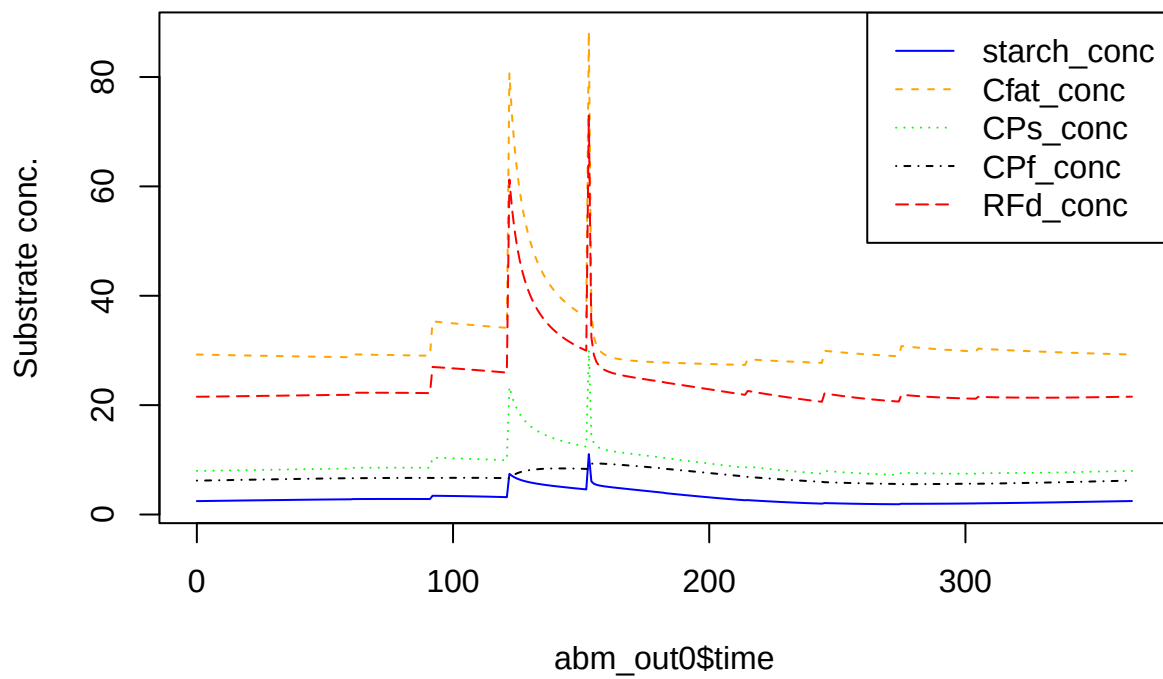
```
matplot(
  abm_out0$time,
  abm_out0[, nn <- grp_pars$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



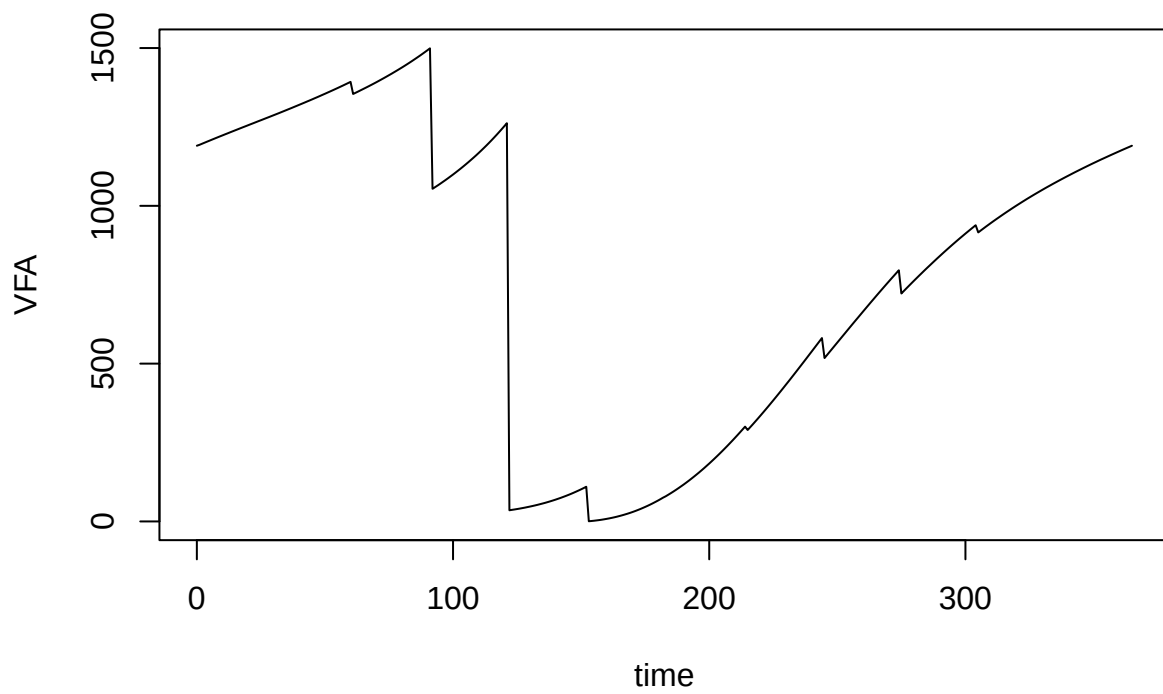
```
matplot(
  abm_out0$time,
  abm_out0[, nn <- paste0(grp_pars$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



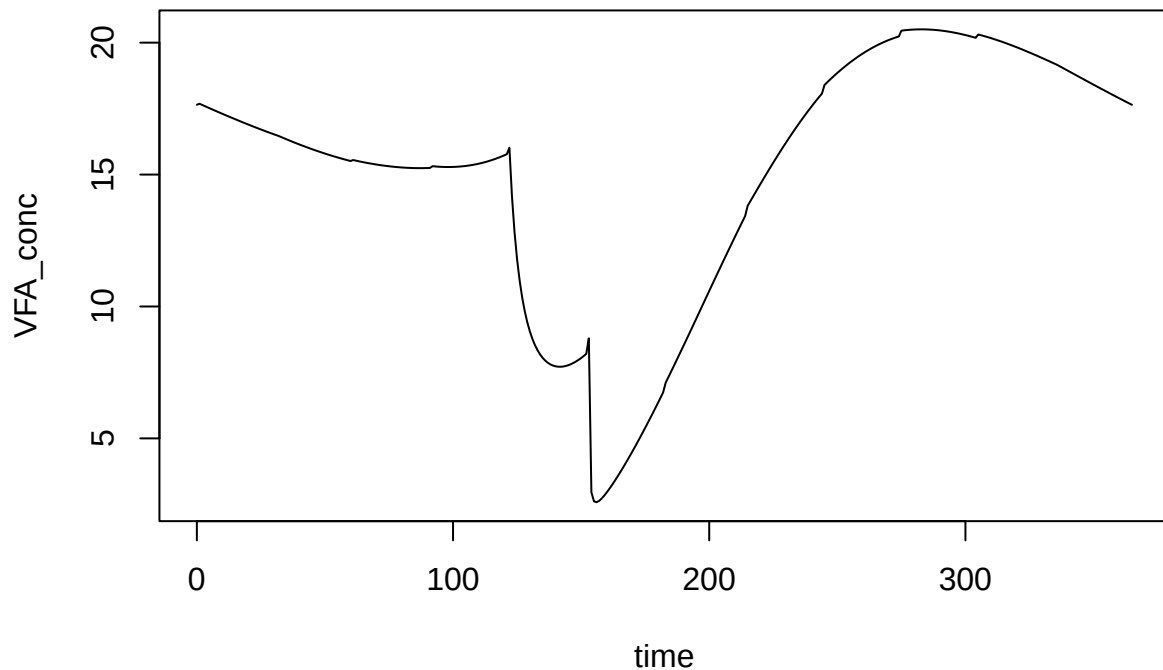
```
matplot(
  abm_out0$time,
  abm_out0[, nn <- paste0(c('starch', 'Cfat', 'CPs', 'CPf', 'RFd'), '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



```
plot(VFA ~ time, data = abm_out0, type = 'l')
```



```
plot(VFA_conc ~ time, data = abm_out0, type = 'l')
```



Individual substrates

Starch first

```
man_pars_starch <- list(conc_fresh = list(sulfide = 0.01, urea = 3.17, sulfate = 0.01, TAN = 0.0, starch = 10.0,
VFA = 0, xa_dead = 0, Cfat = 0, CPs = 0, CPf = 0, RFd = 0, iNDF = 0,
ash = 0), pH = 7, dens = 1000)
```

```
abm_out_starch <- abm(
  days = 100,
  man_pars = man_pars_starch,
  mng_pars = mng_pars,
  arrh_pars = arrh_pars,
  grp_pars = grp_pars0,
  add_pars = list(evap = 0, rain = 0, COD_conv.CH4 = 5.32),
  resp = FALSE,
  pH_inhib_overrule = TRUE,
  startup = 0
)
```

```
## Warning in abm(days = 100, man_pars = man_pars_starch, mng_pars = mng_pars, : Maximum slurry mass exceeded
## Check output.
```

```
## arguments overwritten by slurry_mass: slurry_prod_rate, empty_int, resid_depth, wash_water, wash_int
```

```
## rain = 0 kg/m2/day
```

```
## evaporation = 0 kg/m2/day
```

```
abm_out_starch$sum <- abm_out_starch$starch_conc + abm_out_starch$VFA_conc  
abm_out_starch[1:40, c('time', 'starch_conc', 'VFA_conc', 'sum')]
```

##	time	starch_conc	VFA_conc	sum
## 1	0	1.0000000	0.000000000	1.000000
## 2	1	0.9966588	0.003341305	1.000000
## 3	2	0.9933970	0.006603122	1.000000
## 4	3	0.9902043	0.009795984	1.000000
## 5	4	0.9870789	0.012921406	1.000000
## 6	5	0.9840195	0.015980867	1.000000
## 7	6	0.9810247	0.018975805	1.000000
## 8	7	0.9780929	0.021907623	1.000001
## 9	8	0.9752196	0.024780992	1.000001
## 10	9	0.9724000	0.027600648	1.000001
## 11	10	0.9696331	0.030367652	1.000001
## 12	11	0.9669178	0.033083036	1.000001
## 13	12	0.9642531	0.035747809	1.000001
## 14	13	0.9616380	0.038362955	1.000001
## 15	14	0.9590715	0.040929500	1.000001
## 16	15	0.9565497	0.043451398	1.000001
## 17	16	0.9540681	0.045933029	1.000001
## 18	17	0.9516261	0.048375135	1.000001
## 19	18	0.9492229	0.050778399	1.000001
## 20	19	0.9468578	0.053143487	1.000001
## 21	20	0.9445303	0.055471053	1.000001
## 22	21	0.9422397	0.057761764	1.000001
## 23	22	0.9399824	0.060019100	1.000002
## 24	23	0.9377543	0.062247250	1.000002
## 25	24	0.9355550	0.064446659	1.000002
## 26	25	0.9333839	0.066617761	1.000002
## 27	26	0.9312407	0.068760977	1.000002
## 28	27	0.9291250	0.070876724	1.000002
## 29	28	0.9270366	0.072965193	1.000002
## 30	29	0.9249720	0.075029920	1.000002
## 31	30	0.9229272	0.077074750	1.000002
## 32	31	0.9209021	0.079099926	1.000002
## 33	32	0.9188963	0.081105707	1.000002
## 34	33	0.9169520	0.083050056	1.000002
## 35	34	0.9150269	0.084975225	1.000002
## 36	35	0.9131207	0.086881487	1.000002
## 37	36	0.9112306	0.088771614	1.000002
## 38	37	0.9093526	0.090649646	1.000002
## 39	38	0.9074866	0.092515716	1.000002
## 40	39	0.9056324	0.094369980	1.000002